

Recall: Inclusion–Exclusion & Bonferroni

Given sets $A_1, \dots, A_n$ in a universe S , the **inclusion–exclusion principle** states

$$\left| \bigcup_{i=1}^n A_i \right| = \sum_{k=1}^n (-1)^{k-1} \sum_{\substack{I \subseteq \{1, \dots, n\} \\ |I|=k}} \left| \bigcap_{i \in I} A_i \right|.$$

The **Bonferroni inequalities** say: truncating the right-hand side at the first q terms gives

- an *upper bound* on $|\bigcup A_i|$ when q is even, and
- a *lower bound* on $|\bigcup A_i|$ when q is odd.

Practice: Textbook Exercise 3.8.2 — Prove the Bonferroni inequalities

Prove the Bonferroni inequalities. If you cannot handle the general case, try at least the cases $q = 1$ and $q = 2$.

Remarks

Strategy. For each element x in the universe, let $c(x) = \#\{i : x \in A_i\}$. The contribution of x to the truncated sum at level q is $S_q(c(x)) = \sum_{k=1}^q (-1)^{k-1} \binom{c(x)}{k}$. The problem reduces to showing: for every integer $c \geq 1$,

$$S_q(c) \geq 1 \text{ if } q \text{ is odd,} \quad S_q(c) \leq 1 \text{ if } q \text{ is even.}$$

A useful identity: $\sum_{k=0}^q (-1)^k \binom{c}{k} = (-1)^q \binom{c-1}{q}$ for $c \geq 1$. (Prove it by induction on q using Pascal's rule.)

Answer: Key identity. We claim that for $c \geq 1$ and $q \geq 1$,

$$\sum_{k=0}^q (-1)^k \binom{c}{k} = (-1)^q \binom{c-1}{q}. \quad (*)$$

Proof by induction on q . Base case $q = 0$: both sides equal 1. Inductive step: assuming $(*)$ for $q - 1$,

$$\sum_{k=0}^q (-1)^k \binom{c}{k} = (-1)^{q-1} \binom{c-1}{q-1} + (-1)^q \binom{c}{q}.$$

Factor out $(-1)^q$: $= (-1)^q \left[\binom{c}{q} - \binom{c-1}{q-1} \right] = (-1)^q \binom{c-1}{q}$ by Pascal's rule $\binom{c}{q} = \binom{c-1}{q} + \binom{c-1}{q-1}$.

Main proof. Write $S_q(c) = \sum_{k=1}^q (-1)^{k-1} \binom{c}{k} = 1 - \sum_{k=0}^q (-1)^k \binom{c}{k} = 1 - (-1)^q \binom{c-1}{q}$.

- If q is *odd*: $S_q(c) = 1 + \binom{c-1}{q} \geq 1$. ✓
- If q is *even*: $S_q(c) = 1 - \binom{c-1}{q} \leq 1$. ✓

For $c = 0$ (element not in the union): $S_q(0) = 0$ and the element contributes 0 to both sides. ✓

Since every element's contribution to the truncated sum is ≥ 1 (respectively ≤ 1) times its contribution to $|\bigcup A_i|$, summing over all elements gives the Bonferroni inequalities.

Explicit cases. $q = 1$ (odd, lower bound): $\sum_{i=1}^n |A_i| \geq |\bigcup_{i=1}^n A_i|$ — this is the *union bound* (immediate from overcounting).

$q = 2$ (even, upper bound): $\sum_i |A_i| - \sum_{i < j} |A_i \cap A_j| \leq |\bigcup_i A_i|$. □

How many orderings (permutations) of the 16 letters

A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P

are there such that we *cannot* obtain any of the words **BAD**, **DEAF**, **APE** by crossing out some letters (i.e. as a *subsequence*)?

What if we also forbid **LEADING**?

Remarks

Key observation. A word $w_1 w_2 \cdots w_k$ appears as a subsequence of a permutation iff the letters $w_1, w_2, \dots, w_k$ appear in that relative order. Among all $k!$ orderings of those letters, exactly one gives the forbidden pattern. For intersections, identify the combined partial-order constraints on the involved letters and count *linear extensions* of the resulting poset. Watch out for pairs of constraints that *contradict* each other (e.g. one requires A before D , another requires D before A).

Answer: Let A_1, A_2, A_3 be the sets of permutations containing BAD, DEAF, APE as subsequences, respectively. We want $16! - |A_1 \cup A_2 \cup A_3|$.

Individual sizes. Each word imposes a specific relative order on its letters; the probability that a random permutation respects that order is $1/(\text{word length})!$:

$$|A_1| = \frac{16!}{3!} = \frac{16!}{6}, \quad |A_2| = \frac{16!}{4!} = \frac{16!}{24}, \quad |A_3| = \frac{16!}{3!} = \frac{16!}{6}.$$

Pairwise intersections.

- $A_1 \cap A_2$: BAD requires $A < D$; DEAF requires $D < E < A$, hence $D < A$. Contradiction $\implies |A_1 \cap A_2| = 0$.
- $A_2 \cap A_3$: DEAF requires $E < A$; APE requires $A < P < E$, hence $A < E$. Contradiction $\implies |A_2 \cap A_3| = 0$.
- $A_1 \cap A_3$: BAD requires $B < A < D$; APE requires $A < P < E$. Combined on $\{A, B, D, P, E\}$: $B < A$, $A < D$, $A < P$, $P < E$ — compatible!

Count linear extensions of the poset $B < A < D$, $A < P < E$ (a tree with root A): B must precede A ; after A , the elements D and the chain $P < E$ can interleave. Enumeration: B, A, D, P, E ; B, A, P, D, E ; B, A, P, E, D . That is 3 linear extensions out of $5! = 120$ orderings.

$$|A_1 \cap A_3| = \frac{3}{5!} \cdot 16! = \frac{16!}{40}.$$

Triple intersection: $A_1 \cap A_2 = \emptyset$ already forces $|A_1 \cap A_2 \cap A_3| = 0$.

Inclusion–exclusion.

$$|A_1 \cup A_2 \cup A_3| = 16! \left(\frac{1}{6} + \frac{1}{24} + \frac{1}{6} - \frac{1}{40} \right) = 16! \cdot \frac{20 + 5 + 20 - 3}{120} = \frac{42}{120} \cdot 16! = \frac{7}{20} \cdot 16!.$$

Answer (without LEADING) = $16! - \frac{7}{20} \cdot 16! = \frac{13}{20} \cdot 16!$.

Adding the constraint LEADING. Let A_4 = permutations containing LEADING as a subsequence. $|A_4| = 16!/7! = 16!/5040$.

Intersections with A_4 :

- $A_2 \cap A_4$: DEAF needs $D < E$; LEADING needs $L < E < A < D$, i.e. $E < D$. Contradiction $\implies 0$.
- $A_3 \cap A_4$: APE needs $A < E$; LEADING needs $E < A$. Contradiction $\implies 0$.
- $A_1 \cap A_4$: BAD needs $B < A < D$; LEADING needs $L < E < A < D < I < N < G$. Combined on 8 letters $\{A, B, D, E, G, I, L, N\}$: chain $L < E < A < D < I < N < G$ plus $B < A$. B can be inserted before the chain starts at A : positions before L , between L and E , or between E and A — giving 3 linear extensions out of $8!$.

$$|A_1 \cap A_4| = \frac{3}{8!} \cdot 16! = \frac{16!}{13440}.$$

All higher intersections involving A_4 vanish (each contains either $A_2 \cap A_4 = \emptyset$ or $A_3 \cap A_4 = \emptyset$, and $A_1 \cap A_2 = \emptyset$).

Final inclusion–exclusion (writing $D = 8! = 40320$ as common denominator):

$$\begin{aligned} |A_1 \cup \dots \cup A_4| &= 16! \left(\frac{1}{6} + \frac{1}{24} + \frac{1}{6} + \frac{1}{5040} - \frac{1}{40} - \frac{1}{13440} \right) \\ &= \frac{16!}{40320} (6720 + 1680 + 6720 + 8 - 1008 - 3) = \frac{14117}{40320} \cdot 16!. \end{aligned}$$

$\text{Answer (also forbidding LEADING)} = \frac{40320 - 14117}{40320} \cdot 16! = \frac{26203}{40320} \cdot 16!.$
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